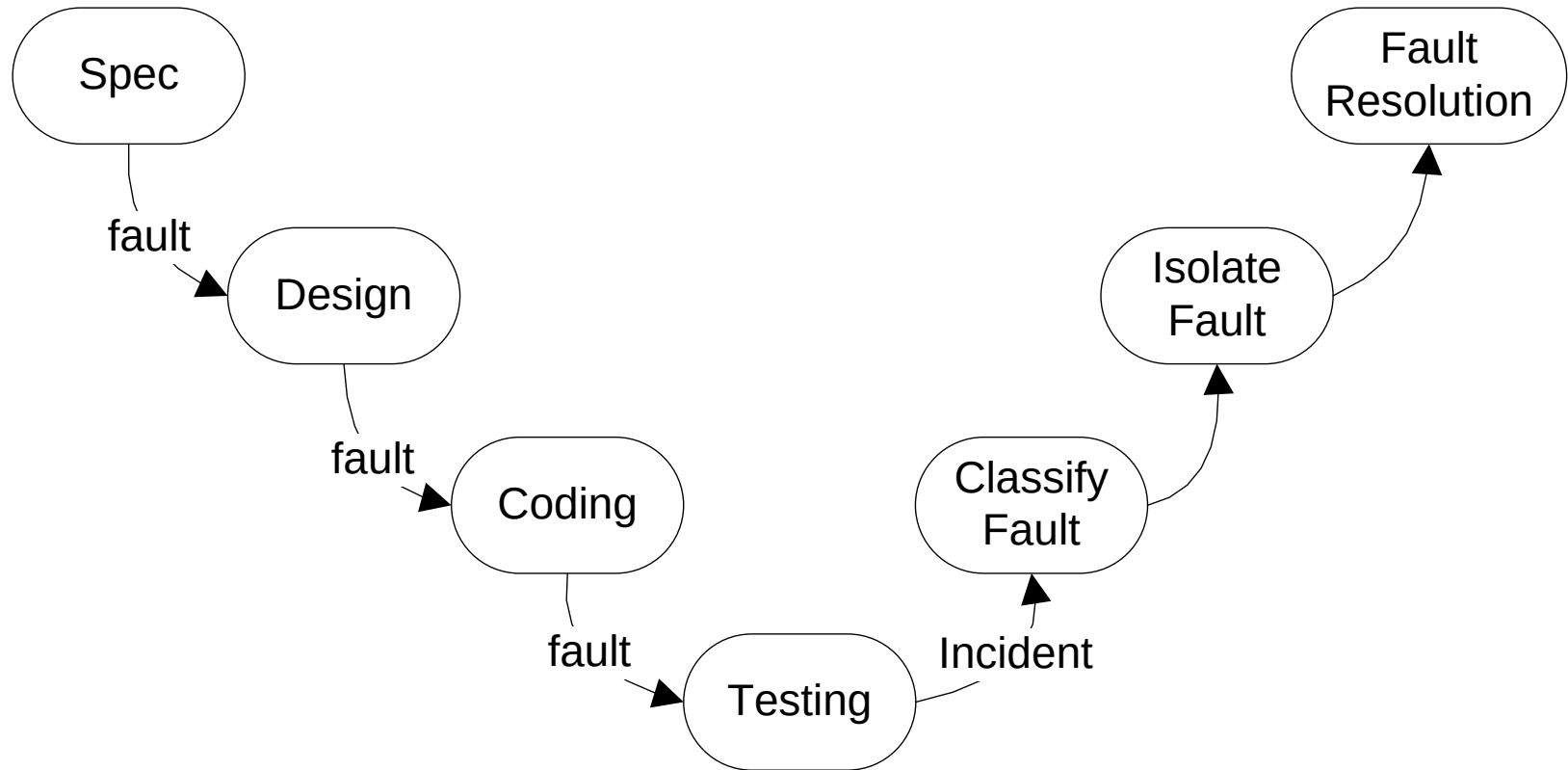


Chapter 1

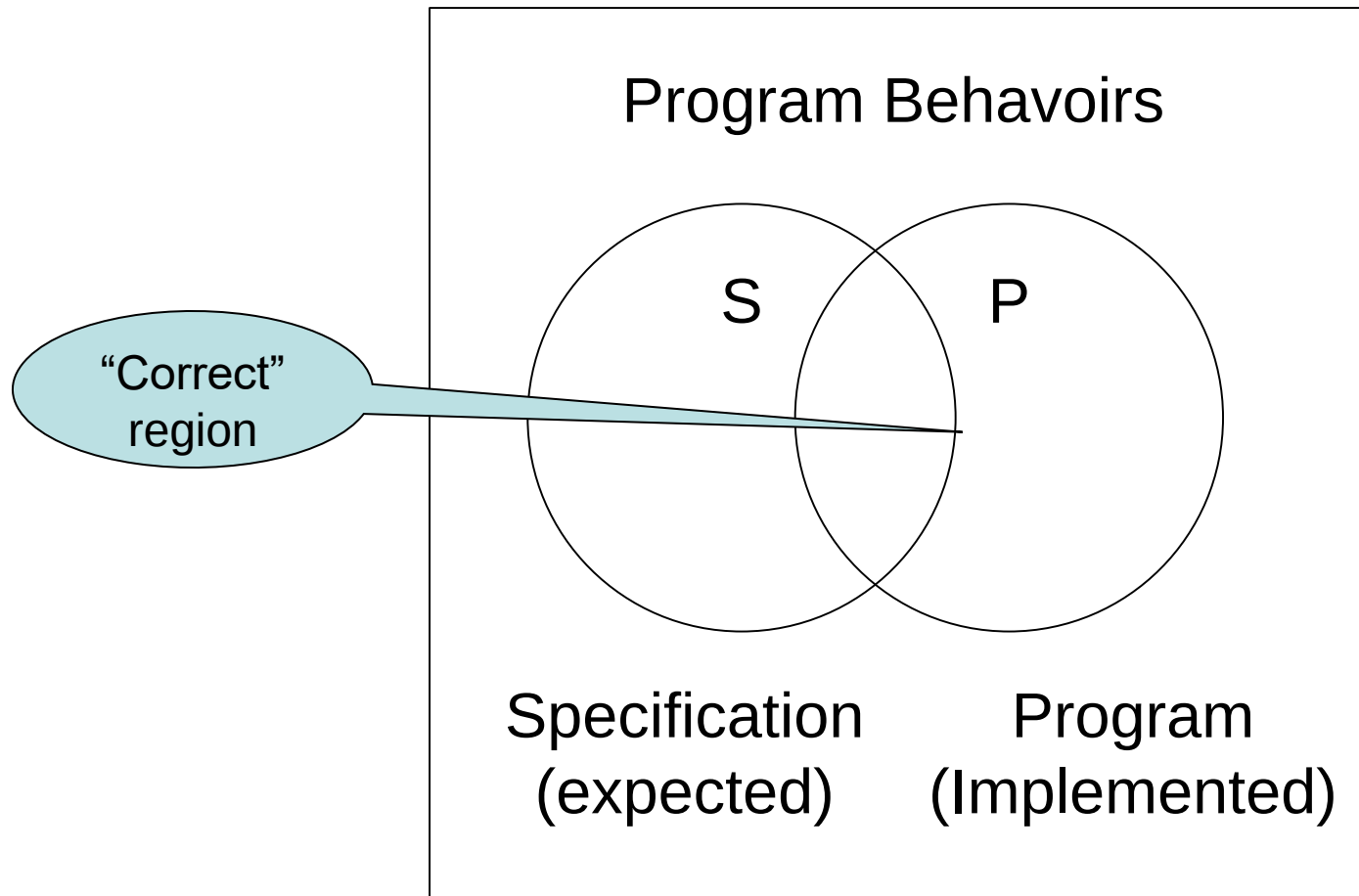
Introduction



Errors, Faults, Failures and Incidents



Universe of Discourse—Program Behaviors

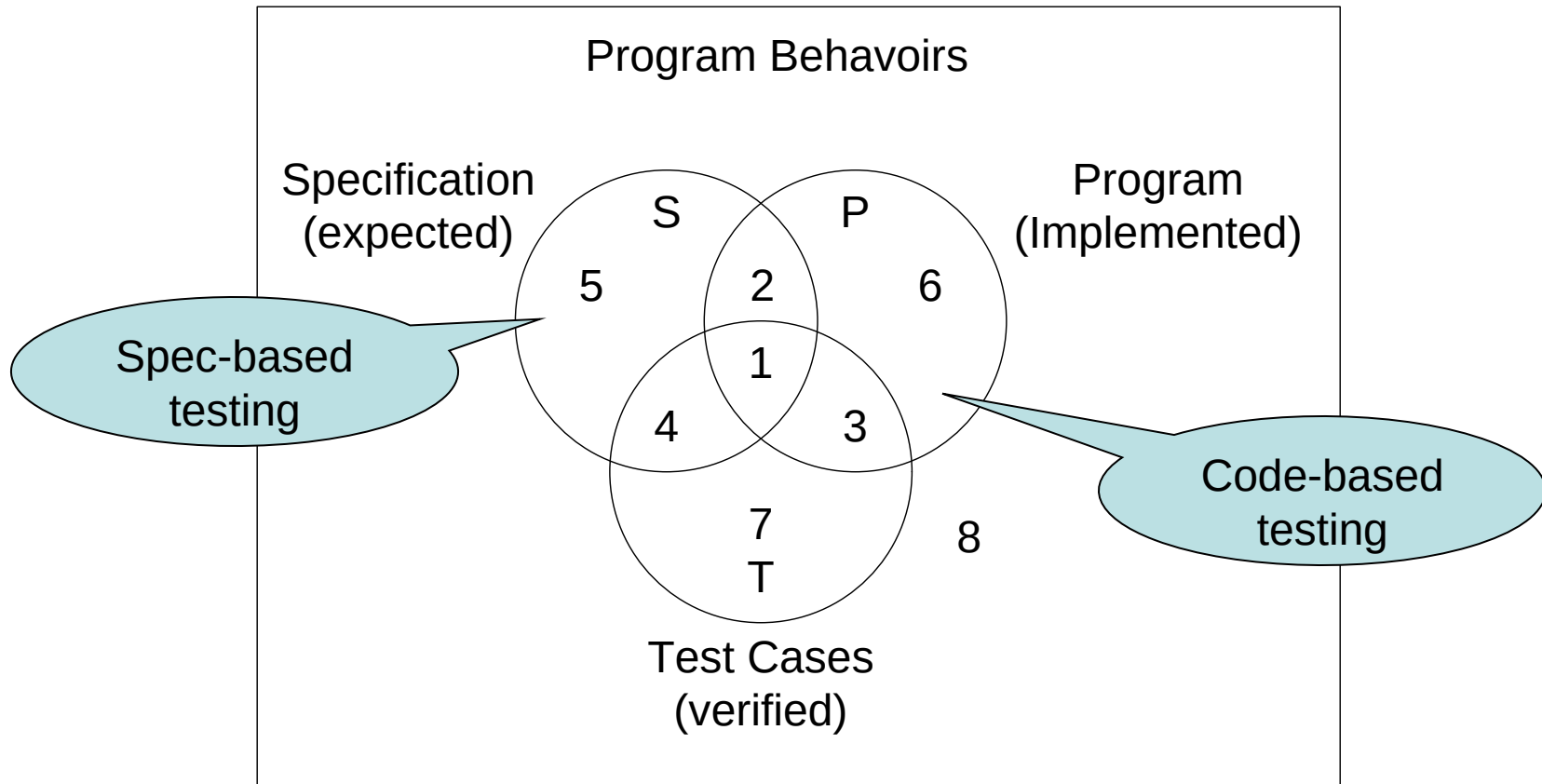


About Correctness...

- Impossible to demonstrate
- A term from “classical” computer science
 - “proofs” derived from code
 - Not derived from specification
 - Can only prove that the code does what it does!
- Better viewpoint: a relative term—program P is correct with respect to specification S.
- Bottom Line: do the specification and the program meet the customer/user's expectations?

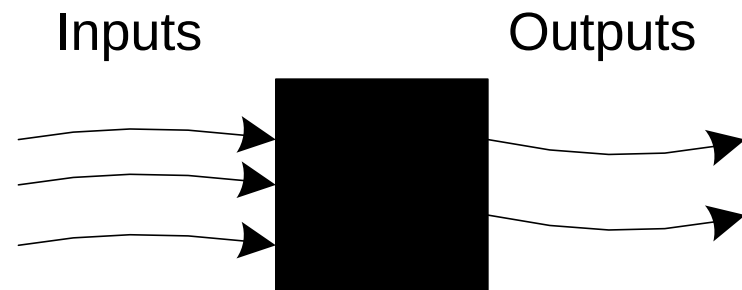


Testing and Program Behaviors



Synonyms—Metaphors gone wild

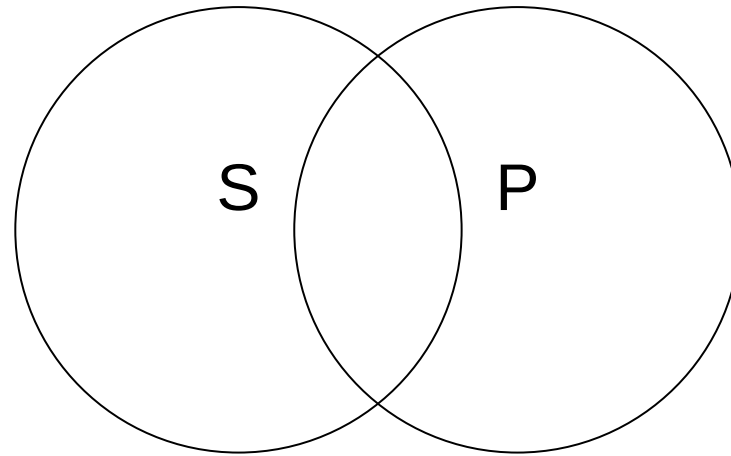
- Specification-based testing
 - Functional testing
 - Black box testing
- Code-based testing
 - Structural testing
 - White box testing
 - Glass box testing
 - Clear box testing



An engineer's Black Box: a system understood in terms of its inputs and outputs



Program Behaviours

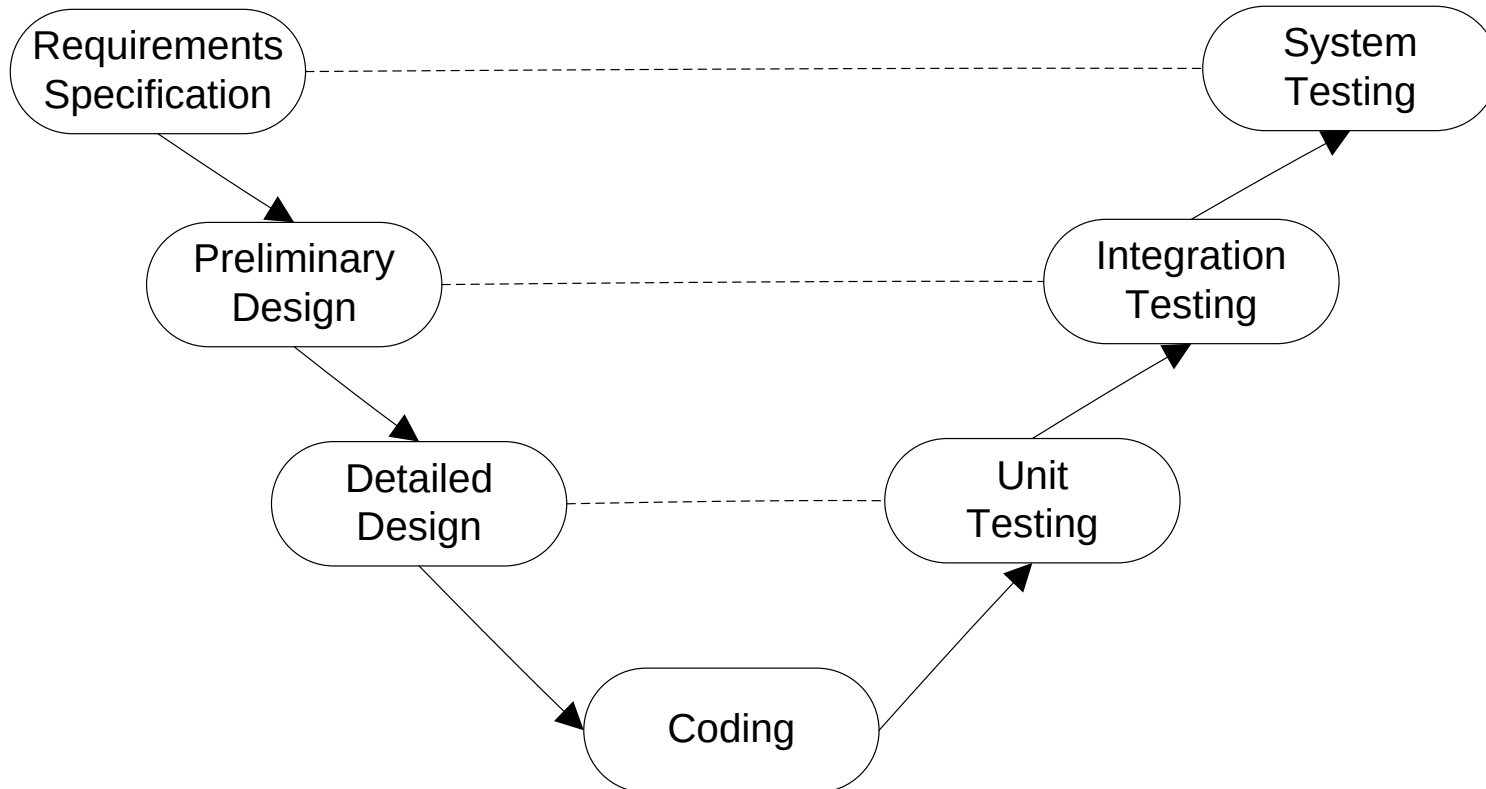


Spec-based
Functional
Black Box
(establishes confidence)

Code-based
Structural
White/Clear Box
(seeks faults)



Levels of Testing (as seen in the V-model)



Content of a Test Case

- Test case identifier (usually a short name for test management purposes)
- Name
- Purpose (e.g., a business rule)
- Pre-conditions (if any)
- Inputs
- Expected outputs
- Observed outputs
- Pass/fail?



Testing as an Experiment

- With respect to the Error, Fault, Failure, Incident framework, a test case is an experiment that...
 - is designed to anticipate an error
 - that is realized as a fault
 - that causes a failure
 - and is recognized by comparing expected and observed outputs.
- In this framework, test cases are intended to reveal incidents.



How Might a Test Case Fail?

- False positive?
- False negative?
- What do these really mean?
- Exercise
 - Analyze false positive, false negative
 - Suggestion: use a decision table with conditions:
 - Expected output correct?
 - Observed output correct? (potential circularity here)
- Compare with common medical usage



